

**Project Proposal**

**Clark University School of Business**

**STAT 4600**

**Intermediary Statistical Modeling for Analytics**

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(Logo created with Canva.com)

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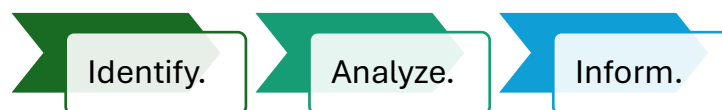
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## **I. Attribution Statement**

All the code represented in this proposal was taken from the following sources:

- a. Grolemund, Garrett, and Hadley Wickham. *R for Data Science*. O'Reilly Media, 2017.
- b. Long, JD, and Paul Teetor. *R Cookbook*. 2nd ed., O'Reilly Media, 2019. ISBN: 978-1-492-04068-2.

## **II. Significance.**

### ***A. The Research Topic***

The focus of this research is to test whether measures of Emotional Intelligence be accurate predictors of academic success in graduate school, by comparing measures of student success (standardized test scores, GPA) to measures of Emotional Intelligence.

### ***B. The Research Problem.***

The primary issue with traditional predictors of academic performance is that they are limited in assessing a student's true potential in a rigorous academic setting. Traditional measures of academic performance include standardized criteria such as test scores, IQ, grade point average (GPA), SATs, and GREs, which measure cognitive ability. Recent studies have shown that these measures are not accurate when evaluating academic performance (Haynes, 2023; MacCann, 2019). A 2008 study evaluating the SAT scores for 50,000 students found a correlation of 0.41, accounting for 16% of the variance reported in college GPAs (Haynes, 2023; Kobrin et al., 2008). Such scores provide a limited scope of a student's aptitude, excluding a student's socio-economic circumstances, inequalities in access to educational opportunities, or the behavioral/emotional aspects of the student's personality (Haynes, 2023; Sepehrian, 2012).

Over the last two decades, Emotional Intelligence (EI) skills, encompassing constructs such as self-awareness, empathy, stress management, and emotional self-regulation, have been proposed as an alternative to traditional cognitive markers in predicting academic success (Haynes, 2023; MacCann et al., 2019; Sepehrian, 2012). This proposal posits that a broader understanding of what drives academic performance can be better understood by exploring EI as a potential predictive marker for academic success. Using data acquired from the first comprehensive meta-analysis generated by McCann et.al. of the University of New South Wales

in 2019, exploring the relationship between standardized scores, emotional intelligence, and academic success will be analyzed using analytical models and machine learning.

The focus of this analysis will be the cohort of 553 graduate student studies present in the available dataset consisting of data collected between 2000 and 2019. It is the goal of this analysis to measure if emotional intelligence is an accurate predictive marker for academic performance in graduate students, compared to standardized test scores or GPA. The graduate cohort consists of students in graduate school, defined as master's, medical, or PhD students. This cohort represents a global geographical region (Table 1). Of the studies represented in the trimmed data: 52.04% came from journal articles, 40.77% came from dissertations, 6.95% came from unpublished data, and 0.24% came from other sources. The original data was trimmed to equal education level 2 (graduate) and by Ei measure (1.1 (MSCEIT) and 3.1 (EQI)). A full description of the trimming process and the logic behind it can be found under the methods section. Based on the content of the data it was decided to focus on two EI streams: ability and mixed.

| Countries Represented in the Data Set (A-Z) |                          |          |                      |
|---------------------------------------------|--------------------------|----------|----------------------|
| Australia                                   | Bahrain                  | Barbados | Belgium              |
| Canada                                      | China                    | Cyprus   | Europe               |
| India                                       | Iran                     | Israel   | Italy                |
| Malaysia                                    | Nigeria                  | Pakistan | Romania              |
| Saudi Arabia                                | Spain                    | Turkey   | United Arab Emirates |
| United Kingdom                              | United States of America |          |                      |

**Table 1:** Overview of the distribution of the following variables from the trimmed dataset used for the study: number of columns, number of rows, publication type, education level, EI stream, and EI measure. Distributions were calculated in R. (MacCann et.al., 2015)

The distribution of the EI streams in the trimmed data set were as follows: ability EI (48.68 %) and mixed EI (51.32 %) (Table 2). Ability EI has test-takers demonstrate knowledge of or evaluate emotion-based information (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). These tests evaluate EI as a cognitive ability like verbal acuity (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). The representative

measure for ability EI is the MSCEIT (The Mayer-Salavoy-Caruso Emotional Intelligence Test) (Table 2). The MSCEIT measures the “four branches” of the Mayer-Salavoy-Caruso model, including: the test takers ability to observe emotions, produce emotion, recognize emotions, and

| Variable         | Sub-category Proportion (%) |                        |                 |           | Total |
|------------------|-----------------------------|------------------------|-----------------|-----------|-------|
| Columns          |                             |                        |                 |           | 20    |
| Rows             |                             |                        |                 |           | 553   |
|                  | 0 (Journal Articles)        | 1 (Dissertations)      | 3 (Unpublished) | 4 (Other) |       |
| Pub_type         | 0.5204                      | 0.4077                 | 0.0695          | 0.0024    | 1     |
|                  |                             |                        | 2 (Graduate)    |           |       |
| ed_level1        |                             |                        | 1               |           | 1     |
|                  | 1 (Ability EI)              | 2 (Self-rated EI)      | 3 (Mixed EI)    |           |       |
| ei_stream        | 0.4868                      |                        | 0.5132          |           | 1     |
|                  | 0 (GPA)                     | 1 (Standardized Score) |                 |           |       |
| achievement_type | 0.8321                      | 0.1573                 |                 |           | 1     |

**Table 2:** Overview of the distribution of the following variables from the trimmed dataset used for the study: number of columns, number of rows, publication type, education level, EI stream, and EI measure. Distributions were calculated in R. (MacCann et.al., 2015)

control emotions (MSCEIT, 2024). Mixed EI, measured by EQI (Emotional Quotient Inventory), encompasses a mixture of personality and behavior assessments, which evaluate emotion-related outcomes such as performance in the workplace (Table 2) (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). The EQI test specifically measures how EI (social and emotional functioning) affects individuals in the workplace, in relation to teamwork, managing stress, and conflict management (EQI, 2024). Since these tests measure different EI constructs, they are only weakly correlated (O'Connor et. al., 2015; MacCann et.al., 2015). The results of the MSCEIT and EQI are then compared to given cognitive measures of success, either GPA scores represented in 83.21% of the data or standardized test scores represented by 16.79% of the data (Table 2) (MacCann et.al., 2015). These correlations were used to calculate the effect size of each study, which was then converted into an  $r^2$  value.

### C. Study Question and Study Aim

Using a regression analysis and a decision tree analysis, this study aims to demonstrate that there is a correlation between academic achievement in the form of a GPA score or standardized test score and one of two EI streams (ability EI or mixed EI). The measures of academic achievement will represent the dependent variable in this study and each EI measure will represent the independent variables. Since the correlation coefficients for this relationship have already been calculated, the resulting  $r^2$  values will be used to calculate the machine learning model used. This aim will address the following research question: can measures of Emotional Intelligence be

accurate predictors of academic success in graduate school, by comparing measures of student success (standardized test scores, GPA) to measures of Emotional Intelligence? By positing this question, this study hopes to inform future graduate enrollment policies, paving the way for the use of more comprehensive measures of potential student success in higher education, rather than a focus on mere measures of cognitive ability.

#### **D. *Importance of the Research Problem.***

This research is important because it will address the limitations of traditional measures of academic potential, while highlighting the utility of EI measures in admissions criteria. GRE and GMAT tests traditionally evaluate general cognitive ability, including verbal, quantitative, and, writing skills (Moneta-Koehler et.al., 2017). These exams are admission requirements for many graduate programs. The Education Testing Service (ETS), which creates and proctors the GRE examination, cautions against using the test as a valid predictor of academic success, as it overlooks other skills vital to succeeding in a rigorous academic program (Haynes, 2023; Moneta-Koehler et al., 2017). EI encompasses constructs such as perseverance, emotional resilience, and adaptability, which are essential to navigating the challenges of an academic environment (Haynes, 2023; Moneta-Koehler et al., 2017; Romanelli et al., 2006). The importance placed on standardized testing scores and GPA has been questioned in many studies and institutions, resulting in approximately 50% of science-based graduate programs dropping their GRE admissions requirement by 2020 (Langin, 2019). Considering this statistic, EI measurements could be a more accurate replacement for the GRE. In graduate-level education, skills such as coping with stress, people management skills, and time management are vital tools (MacCann et.al., 2019; Moneta-Koehler et.al., 2017; Romanelli et. al., 2006). Based on this preliminary evidence, can emotional intelligence be an accurate metric for academic performance compared to standardized tests?

#### **E. *Literature Review***

##### ***EI and It's Future Applications***

Several studies have highlighted the potential benefits of EI-based assessments in not only improving the well-being of students and promoting future job satisfaction but also improving admissions processes and, therefore, retention rates in graduate programs. Global graduate degree retention rates have remained around 75% over the last decade (NCES, 2022).

Admissions decisions that include EI evaluations will provide a more well-rounded approach to identifying potential candidates for rigorous academic programs (MacCann et.al., 2019; Moneta-Koehler et.al., 2017; Romanelli et. al., 2006). Using EI-based candidate selection could lead to higher retention rates in graduate programs, with increased chances of graduates excelling both academically and professionally (MacCann et.al., 2019; Moneta-Koehler et.al., 2017; Romanelli et. al., 2006). Many colleges and universities have begun emphasizing mental health awareness and stress management when providing supportive on-campus programming for students.

Stress is present every day among students in graduate school. Although graduate students encounter stress, anxiety, and isolation in their academics, they do not always know how to seek help. In a study measuring the stress levels among 2,683 master's, doctoral, and professional graduate students, 75% of the sample population reported moderate to severe stress levels (Allen et.al., 2022). EI-based programming could include stress management, emotional regulation, and social skills training classes (MacCann et. al., 2019). Overall, student well-being can be improved by these measures. These statistics highlight the need for more studies focusing on the potential impact of EI on graduate academics, as the benefits of EI can extend beyond the classroom, influencing a student's personal and professional life.

The conclusions drawn by this work are not limited to academics but can also affect career development. Institutions must emphasize EI development to promote lifelong success within their student body. Several studies have shown that high EI correlates with job satisfaction and performance, with one study reporting a correlation of (0.19,  $p < 0.001$ ), indicating a significant positive relationship (MacCann et.al., 2019; Sorio-Vasquez et. al., 2023). Dubai workplace research also showed that EI plays a significant role in leadership skills, working in a team environment, communication skills, and conflict management (Nair et.al., 2016). By exploring the association between EI and academic performance, this study indicates that students who acquire qualities of EI are more successful in the job market (Nair et.al., 2016).

## ***Conclusions***

This research seeks to expand on the existing literature by presenting quantitative findings on the correlation between EI and academic performance. It seeks to enrich the field of psychology and its application to educational approaches. The conclusions reached in this study could inform education policy and promote change to the currently accepted paradigm within our education

system (Zins & Elias, 2006). Broadening our understanding of how we learn can lead to a more comprehensive approach to student success across academic, personal, and professional domains (Zins & Elias, 2006; Durlak et al., 2011). This expanded perspective underscores the importance of preparing students to thrive as successful individuals capable of facing real-world challenges.

### ***Review of the Gaps within Current Literature***

- 1) Ashkanasy, Neal M., and Catherine S. Daus. “Rumors of the Death of Emotional Intelligence in Organizational Behavior Are Vastly Exaggerated.” *Journal of Organizational Behavior*, vol. 26, no. 4, 2005, pp. 441–52. *Wiley Online Library*, <https://doi.org/10.1002/job.320>.

#### **a) Research Gaps:**

##### *i) Limitations of Methodology.*

- (1) It is not clear what research design has been adopted for this study.
- (2) There is no explanation of how the four-branch abilities test was designed and tested.
- (3) There are few details on the number of participants, their backgrounds, and the research design.

##### *ii) Theoretical Limitations.*

- (1) There is little explanation of the Mayer-Salovey model of emotional intelligence.
  - (a) Little explanation is made about how different the three research streams are.
  - (b) There is no clear-cut definition of emotional intelligence.
  - (c) The abstract was presented as a point/counterpoint without much background.
  - (d) Scientific advances in emotion-based research are not discussed.
  - (e) Specific examples of how emotional intelligence can impact organizational behavior are not given.

##### *iii) Measurement Instrument Limitations.*

- (1) Varied types of tests are only briefly mentioned: ability tests versus self-reporting measurements.
- (2) There is no critical analysis of the strengths and weaknesses of the different measurement approaches.
- (3) No evidence was provided to demonstrate the reliability or validity of the tools.

iv) *Contextual Limitations.*

- (1) It only focuses on organizational behavior.
- (2) This study does not extend emotional intelligence into other realms such as education or personal relationships.
- (3) Discussion of use does not extend beyond the organization.

v) *Theoretical Critique Limitations.*

- (1) Does not address any criticisms while fully explaining them.
- (2) There is a lack of explanation for the meaning of a "utopian political agenda."
- (3) The scientific basis for the study of emotional intelligence is weak.

vi) *Future Research Implications.*

- (1) There are no clear suggestions as to what future research should investigate.
- (2) It gives little guidance on how any future research could explore emotional intelligence.
- (3) There was no mention of key themes where future research could focus.

vii) *Gaps in Understanding.*

- (1) It fails to explain how emotional intelligence varies from similar concepts.
- (2) It also does not detail how emotional intelligence differs from social intelligence.
  - (a) The confines of the concept were not discussed at length.
  - (b) The abstract only proves that this is indeed an academic defense of research regarding emotional intelligence; however, it raises several questions and seems to offer a full article rather than research findings.

2) **Allen, Hannah K., et al. "Stress and Burnout among Graduate Students: Moderation by Sleep Duration and Quality." *International Journal of Behavioral Medicine*, vol. 28, no. 1, Feb. 2021, p. 21. *pmc.ncbi.nlm.nih.gov*, <https://doi.org/10.1007/s12529-020-09867-8>.**

**a) Research Gaps:**

i) *Limitations of Methodology.*

- (1) The study is limited to two large, public universities and cannot be generalized to other graduate student populations.
- (2) The sample may not be diverse enough in terms of:
  - (a) Geographic representation.
  - (b) Types of graduate programs.

(c) Institutional contexts.

(d) The cross-sectional design precludes establishing causal relationships.

ii) *Measurement and Analytical Gaps.*

(1) This study does not say how the measurement of stress was calculated.

(2) Specific instruments used for the measurement of sleep duration and quality were not mentioned.

(3) Classifications are necessary for categorization such as "good" and "poor."

(4) There is little effort to search for the mechanism linking sleep to stress and exhaustion but not to cynicism or inefficiency.

(5) No extensive research has been done on potential confounding variables.

iii) *Demographics.*

(1) There is no demographic representation like age group, gender, ethnicity, etc.

(2) There is no research into the group differences of graduate students regarding sleep and stress. There is no specification regarding which master's, doctoral, and professional programs the study focuses on.

iv) *Limitations About Context.*

(1) This article does not include any research on: Sources of stress from specific programs. Differences in stress levels and sleeping habits among different fields of study. Availability of resources from schools. Limited details about the larger issues that cause stress for graduate students.

v) *Implications for Intervention and Support.*

(1) The summary is that sleep habits might improve, yet no explicit guidance or recommendation exists.

(2) There is no mention of practical help or support plans.

(3) Little detail on how universities and students can manage their stress and sleeping problems.

vi) *Theoretical Gaps.*

(1) No theory is developed which can explain how stress, sleep, and burnout occur in relation to each other.

(2) No detailed discussion on the mental or physical mechanism for such associations.

- (3) There is limited discussion regarding the existing literature on graduate student well-being.

vii) *Recommendations for Future Research.*

- (1) The conclusion calls for longitudinal designs while failing to specify what longitudinally needs to be studied; possible research methodologies, and what key variables need to be monitored over time.

viii) *Mental Health Context.*

- (1) There is no deeper exploration or overall assessment of the individual propensity for mental health problems beyond the brief mentioning, apart from the burnout. There is limited consideration of the need for broader mental health support for graduate students. These gaps suggest opportunities for more comprehensive, nuanced research into graduate student stress, sleep, and well-being, including the need for multi-faceted, contextually rich studies.

**3) Haynes, N. “The Limitations of Test Scores and GPAs in College Admissions.”**

***ACADEME BLOG*, 13 July 2023, <https://academeblog.org/2023/07/13/the-limitations-of-test-scores-and-gpas-in-college-admissions/>.**

**a) Research Gaps:**

i) *Narrow Research Limits.*

- (1) The analysis centers on standardized test scores and high school grade point averages, excluding other potential criteria for college admission.
- (2) The samples contain student populations between 23,000 and 50,000, which may not be representative of the even more diverse body of students at large.

ii) *Methodological Constraints.*

- (1) The methods used in the studies referred to are not discussed in any detail.
- (2) It does not go into detail about how the correlation coefficients were calculated or what statistical methods were used exactly.
- (3) It does not closely examine the possible limits or biases in the studies that are under review.

iii) *Absent Viewpoints.*

- (1) The review mainly gives reasons in favor of affirmative action and criticizes standardized testing.

(2) It lacks a comprehensive view since it does not comprise divergent opinions or opposing ideas about college admissions.

(3) The review seems biased with little evidence to show that there is an understanding of what students need or want.

*iv) Inadequate Contextual Analysis.*

(1) The paper discusses issues of injustices in the system but does not probe into more profound causes for these inequalities.

(2) There is little to no mention of socioeconomic factors other than educational resources.

(3) This review is not meant to give an in-depth background about affirmative action and its application in college admissions.

*v) Insufficient Amount of Data.*

(1) Most correlation research emphasizes the first-year college GPA as a major indicator of academic success.

(2) Longitudinal tracking of student success, career paths, or other global measures of educational achievement does not exist.

(3) The review does not explain how different admission practices affect student diversity and long-term fairness in education.

*vi) Few Real-Life Examples.*

(1) The paper makes strong statements about how standardized testing is not enough, but there isn't much data to back this up.

(2) There is little research that compares different admission models and how well they work.

(3) The discussion lacks international comparisons or insights based on other education systems.

*vii) Insufficient Geographic Bounds.*

(1) Many of the studies and examples come from the United States.

(2) No discussion of college admissions and affirmative action in non-US institutions.

*viii) Inadequate focus on Intersectional Analysis.*

(1) While discussing race and educational inequality, the review does not provide enough attention to the topic of intersectionality. Very little is written about how

race interacts with other identities—such as gender, disability, or socioeconomic status—in the process of college admissions. The identified gaps indicate that, though useful in its provision of information on challenges in college admissions and standardized testing, the literature review would have been better served with deeper and more cautious scrutiny of educational fairness and its processes of admission.

4) **Langin, K. (2019). *A wave of graduate programs drops the GRE application requirement*.**

**Science link - <https://www.science.org/content/article/wave-graduate-programs-drop-gre-application-requirement>**

a) **Objective:** The article analyzes the phenomenon of graduate program's exclusion of the GRE, with an emphasis on its flaw and its effect on diversity and inclusion.

b) **Methodology:** Summative self-generated data from interviews with the faculties involved, data on the programs and an examination of general trends in the admission processes.

c) **Research Gaps:**

i) Neither concrete objectives nor guidelines for complicated assessments are given in discussions of holistic admissions evaluations.

ii) **Key Findings:** Criticism of the GRE is ineffective in assessing success, incurs costs, and could be prejudicial to minority candidates.

iii) **Diversity Impact:** Several programs testify to a rise in the number of applications from minorities and women after the elimination of the GRE requirement.

iv) **Challenges:** Controversy remains as to the absence of coherent criterion for measuring the candidate's performance.

d) **Conclusion:** The move from the usage of GRE as a means of assessment expands the possibilities of getting in students of color, but more study is required. Enrollment of students of color will increase due to eliminating the usage of the GRE, but more research is called for in order to understand the changes.

5) **Moneta-Koehler, L., et al. (2017). *The Limitations of the GRE in Predicting Success in Biomedical Graduate School*. PLOS ONE. <https://doi.org/10.1371/journal.pone.0166742>**

- a) **Objective:** The effectiveness of GRE scores in predicting the time of completion, qualifying exams, rate of completion, the research output and grant records of the Ph.D. students in the biomedical graduate programs are also assessed in this study.
- b) **Research Gaps:** Sample drawn from only one institution, which can be a small population to compare with other programs in existence.
- c) **Methodology:** Biomedical GRE scores for admissions and performance were statistically compared to determining the relationship between scores and success of students attending Vanderbilt University.
- d) **Key Findings:** This research indicated that GRE scores are not useful in determining who will succeed in biomedical graduate programs since correlations between GRE scores and graduate outcomes are low and inconsistent with Ph.D. completion, subsequent research productivity or grant acquisition.
- e) **Conclusion:** In this regard, the current study found that the GRE has low validity in predicting performance in biomedical graduate programs, supporting the adoption of comprehensive intake processes.

### III. Data.

#### *A. Type/Source/Content*

In 2019, MacCann et al. of the University of New South Wales wrote the first comprehensive meta-analysis addressing the correlation between EI and academic performance (MacCann, 2019). The original study included the results from 1,276 studies exploring the connection between EI and academic performance (MacCann, 2019). The collected studies represent nine databases: ERIC, Google Scholar, ISI Web of Science, Medline, ProQuest, PsycINFO, PubMed, Science Direct, and Scopus (MacCann, 2019). Papers were included based on the following criteria: presented in English, the measure of EI used were published in a test manual or journal, measures of academic performance were included (GPA, SAT, GRE), presented original data, and reported the sample size (MacCann, 2019). The included studies were published between 1997 and 2019 and represented 27 different countries (MacCann, 2019). A description of each variable used to represent each study is in the next paragraph (Table 1).

#### *B. Description of Variables*

Twenty categorical and quantitative variables that met the inclusion criteria illustrate the statistics for each study: the effect size ( $y_i$ ), variance ( $v_i$ ), the EI construct used, the EI measure used, and the EI stream (Table 3; MacCann, 2019). The categorical variables were present in the studies represented in the meta-analysis. Table 3 outlines a complete list and description of the twenty variables used within the study. The effect size represents the strength of the relationship between the variables and is equal to a Z-score for a standard normal distribution (Kim, 2015; Sullivan and Fein, 2012). An effect size is calculated by measuring the difference in mean between two independent groups and then dividing that value by the standard deviation (Sullivan and Fein, 2012). The more extensive the effect size, the more significant the findings are (Kim, 2015; Sullivan and Fein, 2012). Conversion of effect size to a Pearson's  $r$  correlation is a standard index or statistical measurement of change (Kim, 2015). Pearson's correlation measures association and represents the effect size for the relationship between two variables (Kim, 2015). A Pearson's correlation falls between -1 and 1, with -1 representing a perfect negative relationship, 1 representing a perfect positive relationship, and 0 representing no relationship (Kim, 2015; Sullivan and Fein, 2012). MacCann et al. reported an overall effect of  $p = 0.20$  with a variance estimation of ( $n = 42,529$ ,  $k = 1,246$  from 158 citations) (MacCann, 2019). Next, the stream and measures of EI require further clarification.

### ***C. Description of the EI Measures Present in the Original Data***

Among the variables represented, two categories represent emotional intelligence measures: EI stream and EI measure. EI stream represents three theoretical types of EI, which MacCann et al. refer to as ability EI, self-rated EI, and mixed EI (MacCann et.al., 2019). Ashkanasy and Daus, 2005, conceptualized these three streams in 2005 based on different Ei measurement methods (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). The first stream, ability EI, has test-takers show knowledge of or evaluate emotion-based (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). These tests evaluate EI as a cognitive ability like verbal acuity (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al.,

| Name             | Description                                                                             | Sub-Category Description                                                                                                                                                      |
|------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| study_id         | Unique ID coding for each study                                                         |                                                                                                                                                                               |
| sample_id        | Unique ID coding for effect size of each study                                          |                                                                                                                                                                               |
| author           | First author                                                                            |                                                                                                                                                                               |
| pub_year         | Year the paper was published                                                            |                                                                                                                                                                               |
| yi               | Effect size represented as R value                                                      |                                                                                                                                                                               |
| vi               | Sampling Variance for the effect size (SE <sup>2</sup> )                                |                                                                                                                                                                               |
| pub_type         | Type of Publication                                                                     | 1: Dissertation<br>2: Conference Proceedings<br>3: Unpublished Data                                                                                                           |
| n                | Sample Size, calculated from the number of participants contributing to the effect size |                                                                                                                                                                               |
| ed_level1        | Education level at the time of data collection                                          | 0: Primary (K-12)<br>1: Secondary (Undergraduate)<br>2: Tertiary (Graduate)<br>3: Mixed                                                                                       |
| ed_level2        | Education level based on the achievement type reported                                  | 0: Primary (K-12)<br>1: Secondary (Undergraduate)<br>2: Tertiary (Graduate)<br>3: Mixed                                                                                       |
| country          | Country where the study was conducted                                                   |                                                                                                                                                                               |
| perc_white       | Percent white participants (USA samples only)                                           |                                                                                                                                                                               |
| age              | Mean age of participants                                                                |                                                                                                                                                                               |
| perc_female      | Percent female participants                                                             |                                                                                                                                                                               |
| ei_stream        | Types of EI measured                                                                    | 1: Ability EI<br>2: Self-rated EI<br>3: Mixed EI                                                                                                                              |
| ei_measure       | EI test used                                                                            | 1.1 = MSCEIT, 1.2 = MEIS, 1.3: DANVA, 1.4: STEU, 1.5: STEM, 2.1 = SUEIT, 2.2: SSRI, 2.3: SREIT, 2.4: TMMS, 2.5 = WLEIS, 3.1 = EQI, 3.2 = TEIQue                               |
| subject          | Subject area measured for academic performance                                          | 0 = general, 1 = verbal/ LA, 2 = math, 3 = science, 4 = social studies, 5 = foreign language, 6 = psychology, 7 = medicine, 8 = engineering, 9 = physical education, 10 = art |
| humanities       | General subject area                                                                    | 0 = general<br>1 = math and sciences<br>2 = humanities and verbal abilities                                                                                                   |
| achievement type | Achievement type                                                                        | 1: Course grade (GPA, semester grade, or school subject mark)<br>2: Standardized test score (SAT or GRE)                                                                      |

**Table 3:** Descriptions of the 20 variables represented in the data set used in the journal article “Emotional Intelligence Predicts Academic Performance: A Meta-Analysis” by MacCann et al., 2019. Table 1 also outlines sub-categories for applicable variables (MacCann et al., 2019).

2015). The second stream, self-rated EI, has test-takers respond to a series of statements and self-reports using a rating scale based on those statements (Ashkanasy and Daus, 2005; O’Connor et al., 2015; MacCann et.al., 2015). Trait EI measures provide a prediction of actual behavior (Ashkanasy and Daus, 2005; O’Connor et. al., 2015; MacCann et.al., 2015). The third stream,

mixed EI, encompasses a mixture of personality and behavior assessments, which evaluate emotion-related outcomes such as performance in the workplace (Ashkanasy and Daus, 2005; O'Connor et. al., 2015; MacCann et.al., 2015). Since these tests measure different EI constructs, they are only weakly correlated (O'Connor et. al., 2015; MacCann et.al., 2015).

Next, twelve tests that measure EI can be found within this dataset. The following testing methods are subcategorized by the EI stream that they measure—five tests measure ability EI: MSCEIT, MEIS, DANVA, STEU, and STEM (Table 3; MacCann et.al., 2019). Five tests measure Trait EI: SUEIT, SSRI, SREIT, TMMS, and WLEIS (Table 3; MacCann et.al., 2019). Finally, the two tests measure the mixed EI streams: EQI and TIQue (Table 3; MacCann et.al., 2019). MacCann et al., 2019, fully explain these tests in their manuscript. A more detailed explanation of the tests represented in the trimmed dataset can be found in the “Introduction” under “The Research Problem.” These measures were compared to one of two academic measures, GPA or a standardized test score. It is important to note that the authors outlined limitations related to their sample pool and the results of their study.

#### ***D. Limitations.***

The authors noted several limitations of their study. MacCann et al. disclosed that little evidence supports a causal relationship between EI and academic performance, represented by either a GPA or standardized test score (MacCann et.al., 2019). There was a stronger relationship between ability EI and academic performance in comparison to self-rated EI or mixed EI (MacCann et.al., 2019). The authors also concluded that ability EI was a better predictor of academic performance in the humanities compared to the sciences (MacCann et.al., 2019). Self-rated EI was a better predictor of GPA than performance on standardized tests (MacCann et.al., 2019). The effect size, represented by an  $r$  correlation, ranged from approximately -0.70 to 0.73, with a median of 0.155 and a mean of 0.176 (Table 4; MacCann et.al., 2019). The variance and standard deviation for the effect size were 0.044 and 0.21, respectively. Since the mean and median are close to 0 for the effect size, there may not be a strong correlation between emotional intelligence (all EI streams) and academic achievement represented in this meta-analysis. The studies included are also snapshots in time and based on individual test scores or grades. To our knowledge, longitudinal studies have not been conducted to measure the relationship between EI,

academic performance, and retention in graduate programs. Further studies investigating these relationships could be justified by the results of this study.

#### IV. Methods

##### A. Preliminary steps.

##### Step 1: Descriptive statistics performed on the untrimmed dataset as a baseline.

**Method:** Descriptive statistics were calculated using R software and represented in table format using Excel, where applicable, to provide a baseline for the data before it was trimmed.

Descriptive statistics were calculated for variables deemed necessary for describing related demographic and statistical information. The R data for MacCann 2019 was sourced using the “psymetadata” repository created by Josue Rodriguez on GitHub (Rodriguez, 2020). The

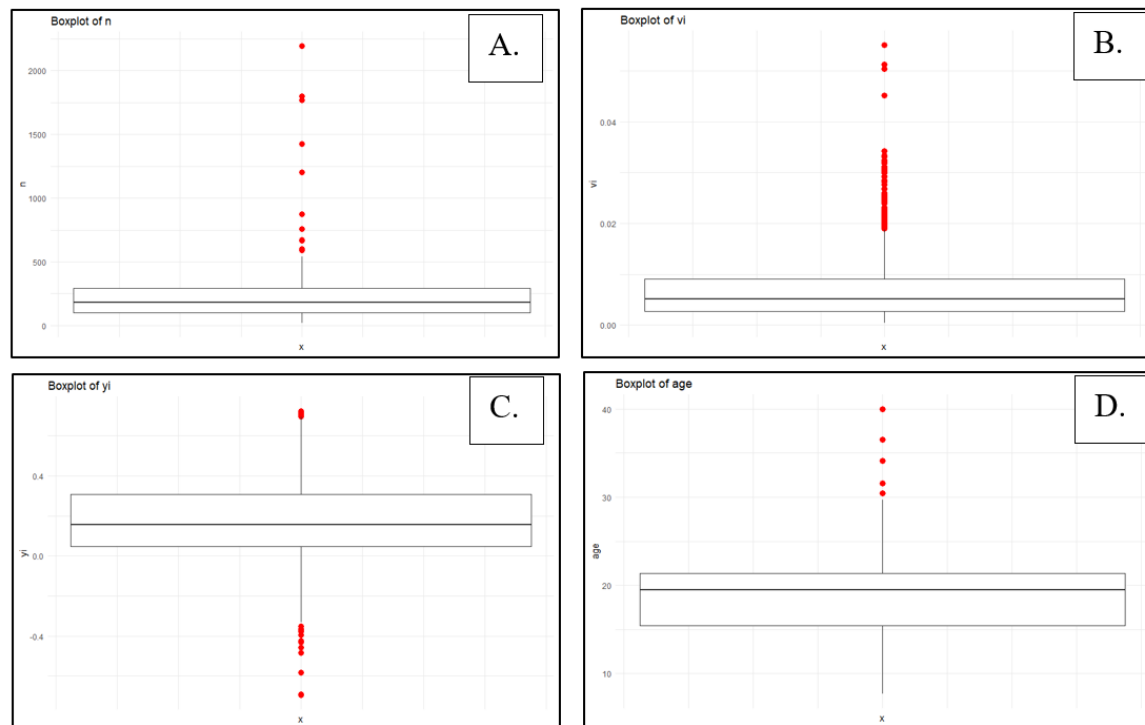
| Variable         | Min       | Max     | Median | Mean   | IQR    | Var      | SD     |
|------------------|-----------|---------|--------|--------|--------|----------|--------|
| pub_year (year)  | 1997      | 2019    | 2010   | 2010   | 8.00   | 19.28    | 4.39   |
| yi (r value)     | -0.69615  | 0.72    | 0.15   | 0.18   | 0.26   | 0.04     | 0.21   |
| vi (SE^2)        | 0.0003938 | 0.06    | 0.01   | 0.01   | 0.01   | 0.00     | 0.01   |
| n (participants) | 18        | 2195.00 | 180.00 | 257.00 | 194.00 | 95577.25 | 309.16 |
| perc_white (%)   | 0         | 100.00  | 52.00  | 54.62  | 45.10  | 763.39   | 27.63  |
| age (years)      | 7.68      | 40.00   | 19.52  | 18.76  | 5.93   | 25.75    | 5.08   |
| perc_female (%)  | 0         | 100.00  | 62.00  | 61.31  | 21.99  | 476.40   | 21.83  |

**Table 4:** Descriptive statistics for the original data set used in MacCann et.al. 2019. Calculations for the following categories are outlined in Tables 2 and 3: publication year, effect size represented by an r correlation value, the variance of the effect size calculated as SE^2, the number of study participants, and demographic information (age, % proportion of the sample that is white, % proportion of the sample that was female) (MacCann, 2019).

publication years range from 1997 to 2019, with a mean and median equal to 2010 (Table 4; MacCann et.al., 2019). The effect size, represented by an r correlation, ranged from -0.70 to 0.73, with a median of 0.155 and a mean of 0.176 (Table 4; MacCann et.al., 2019). The variance and standard deviation for the effect sizer were 0.044 and 0.21, respectively (Table 4; MacCann et.al., 2019). Boxplots were produced to evaluate key variables for outliers (Figure 1A-1D).

**Conclusions:** Since the mean and median are close to 0 for the effect size, there is only a weak positive correlation between emotional intelligence and academic achievement represented in this study. Most of the studiess came from thesis/dissertations and journal articles, 54.9% and 38.2% (Table 3; MacCann et.al., 2019). Demographically, the age range of the study participants ranged from 7.68 to 40 years, with a median of 19.52 years and a mean of 18.76 years (Table 4; MacCann et.al., 2019). The proportion of study participants that were either white or female had a mean of 54.62% and 61.31%, respectively (Table 4; MacCann et.al., 2019). These statistics'

variance and standard deviation were rather significant and skewed by the presence of NA values. Of the populations represented, 7.3% were in primary school, 32.3% were in an undergraduate program, 59.5% were in graduate school, and 0.5% of the studies evaluated mixed populations (Table 3; MacCann et.al., 2019). Of these studies, 30.7% represented the ability EI, 12.8% represented the self-rated EI stream 2, and 56.5% represented the mixed EI stream 3. (Table 3; MacCann et.al., 2019). Finally, the achievement types represented were either GPA or standardized test score-based education level. (Table 3; MacCann et.al., 2019). There were several outliers across four important quantitative variables:  $n$ ,  $vi$ ,  $yi$ , and age. The variable  $n$  represents the sample size, and is right skewed, meaning that the median is closer to the third quartile. This same trend occurs in the boxplot for the variance for each study ( $vi$ ) and the ages



**Figure 1A-1D.:** Boxplots were created in R to evaluate the number of outliers in four quantitative variables:  $n$ ,  $vi$ ,  $yi$ , and age. The variables  $n$ ,  $vi$ , and age are right skewed pushing the median closer to the 3<sup>rd</sup> quartile. The variable  $yi$  is left skewed, pulling the median closer to the 1<sup>st</sup> quartile (MacCann, 2019).

represented. in each study. The effect size converted into an  $r^2$  value for each study,  $yi$ , is left skewed pulling the median closer to the first quartile. This data will be used as a comparison to the trimmed data, which is the focus of this study.

## Step 2: Data Trimming.

**Method 1:** We propose that measures of emotional intelligence are more accurate predictive markers for graduate students' academic performance than standardized test scores. To better answer this question, the data obtained from MacCann et al. was trimmed based on an education level equal to 2, representing the population of graduate students within the dataset (MacCann et.al., 2019). The original data consisted of 1,276 data points, and 746 data points remained after trimming (MacCann et.al., 2019). The variables within the dataset were not affected. Using a proportion table as a quality control measure, the data was checked to make sure that the education level 1 equals 2 (graduate level).

**Code 1:**

```
467 # Filtered McCann 1
468 # 1. Trim Data
469 # filter df by column value, education level 1 = 2 (graduate)
470
471 library(dplyr)
472 filtered_mccann1 <- EI_AP_Data[EI_AP_Data$ed_level1 == 2, ]
473
474 # 746 available samples at the graduate level
475
476 # 2. Summary Statistics
477 nrow(filtered_mccann1)
478 ncol(filtered_mccann1)
479 summary(filtered_mccann1)
480
481 # Quality Check
482 #Education Level 1
483 length(filtered_mccann1$ed_level1)
484 sum(is.na(filtered_mccann1$ed_level1))
485 ed_level1_data_f <- filtered_mccann1$ed_level1[!is.na(filtered_mccann1$ed_level1)]
486 prop_table_f4 <- prop.table(table(ed_level1_data_f)) * 100
487 prop_table_f4 <- round(prop_table_f4, 2)
488 prop_dff4 <- data.frame(Category = names(prop_table_f4), Percentage = prop_table_f4)
489 print(prop_dff4)
```

**Output 1:**

```

> # 1. Trim Data
> # filter df by column value, education level 1 = 2 (graduate)
>
> library(dplyr)
> filtered_mccann1 <- EI_AP_Data[EI_AP_Data$ed_level1 == 2, ]
>
> # 746 available samples at the graduate level
>
> # 2. Summary Statistics
> nrow(filtered_mccann1)
[1] 746
> ncol(filtered_mccann1)
[1] 20
> summary(filtered_mccann1)
  study_id      sample_id      author      pub_year      yi      vi      pub_type      n      ed_level1
Min.   : 5.0   Min.   : 5.00   Length:746   Min.   :1997   Min.   :-0.69319   Min.   :0.0003938   Min.   :0.000   Min.   : 18.0   Min.   :2
1st Qu.: 67.0   1st Qu.: 69.25   Class :character   1st Qu.:2005   1st Qu.: 0.02799   1st Qu.:0.0026885   1st Qu.:0.000   1st Qu.: 85.0   1st Qu.:2
Median :153.0   Median :167.00   Mode  :character   Median :2010   Median : 0.13076   Median :0.0054171   Median :0.000   Median : 163.0   Median :2
Mean   :135.6   Mean   :148.19               Mean :2009   Mean   : 0.14799   Mean   :0.0079933   Mean   :0.756   Mean   : 287.5   Mean   :2
3rd Qu.:195.0   3rd Qu.:220.00               3rd Qu.:2013   3rd Qu.: 0.27544   3rd Qu.:0.0103272   3rd Qu.:1.000   3rd Qu.: 306.0   3rd Qu.:2
Max.   :247.0   Max.   :278.00               Max.   :2019   Max.   : 0.71828   Max.   :0.0551461   Max.   :4.000   Max.   :2195.0   Max.   :2

  ed_level2      country      perc_white      age      perc_female      ei_construct      ei_stream      ei_measure      subject
Min.   :1.000   Length:746   Min.   : 0.00   Min.   : 7.68   Min.   : 0.00   Min.   :0.000   Min.   :1.000   Min.   :1.100   Min.   :0.0000
1st Qu.:2.000   Class :character   1st Qu.: 38.40   1st Qu.:19.90   1st Qu.: 56.97   1st Qu.:0.000   1st Qu.:1.000   1st Qu.:1.100   1st Qu.:0.0000
Median :2.000   Mode  :character   Median : 52.00   Median :21.33   Median : 66.22   Median :3.000   Median :3.000   Median :2.500   Median :0.0000
Mean   :1.862               Mean : 56.73   Mean :22.06   Mean : 63.87   Mean : 3.125   Mean :2.221   Mean :2.253   Mean :0.8587
3rd Qu.:2.000               3rd Qu.: 78.00   3rd Qu.:23.00   3rd Qu.: 73.62   3rd Qu.:5.000   3rd Qu.:3.000   3rd Qu.:3.100   3rd Qu.:0.0000
Max.   :2.000               Max.   :100.00   Max.   :40.00   Max.   :100.00   Max.   :9.000   Max.   :3.000   Max.   :3.300   Max.   :7.0000
NA's   :409               NA's   :1       NA's   :157   NA's   :179   NA's   :136   NA's   :10

  humanities      achievement_type
Min.   :0.0000   Min.   :0.0000
1st Qu.:0.0000   1st Qu.:0.0000
Median :0.0000   Median :0.0000
Mean   :0.1501   Mean   :0.1501
3rd Qu.:0.0000   3rd Qu.:0.0000
Max.   :1.0000   Max.   :1.0000

>
> # Quality Check
> #Education Level 1
> length(filtered_mccann1$ed_level1)
[1] 746
> sum(is.na(filtered_mccann1$ed_level1))
[1] 0
> ed_level1_data_f <- filtered_mccann1$ed_level1[!is.na(filtered_mccann1$ed_level1)]
> prop_table_f4 <- prop.table(table(ed_level1_data_f)) * 100
> prop_table_f4 <- round(prop_table_f4, 2)
> prop_dff4 <- data.frame(Category = names(prop_table_f4), Percentage = prop_table_f4)
> print(prop_dff4)
  Category Percentage.ed_level1_data_f Percentage.Freq
1         2                2                100

```

**Method 2:** The data was then re-evaluated with a focus on the distribution of EI streams and EI measures present in the trimmed data. The distribution of the EI streams within the trimmed data were as follows: ability EI (34.18%), self-rated EI (9.52%), and mixed EI (56.30%). The distribution of the EI measures represented were as follows: 1.1 (MSCEIT) (33.28%), 1.2-2.5 (<5%), 3.1 (EQI) (35.08%), 3.2 (2.62%), and 3.3 (9.84%). Based on this data it was decided to simplify the data by focusing on the two EI measures with the largest distributions, 1.1 (MSCEIT) and 3.1 (EQI). A visual representation of the variable distribution for the trimmed data can be viewed in Table 5. Using a proportion table as a quality control measure, the data was checked to make sure that the EI measures were equal to 1.1 (MSCEIT) or 3.1 (EQI).

**Code 2:**

```

493 # Filter 2
494 # Filter by test 1.1 and 3.1
495
496 # Filter rows where ei_measure is either 1.1 or 3.1
497 filtered_mccann2 <- filtered_mccann1[filtered_mccann1$ei_measure == 1.1 | filtered_mccann1$ei_measure == 3.1, ]
498
499 # Get the number of rows and columns in the filtered data frame
500 nrow(filtered_mccann2) # Number of rows
501 ncol(filtered_mccann2) # Number of columns
502 summary(filtered_mccann2) # Summary statistics
503
504 # Quality Check: EI Measure
505 length(filtered_mccann2$ei_measure)
506 sum(is.na(filtered_mccann2$ei_measure))
507 ei_measure_data_f2 <- filtered_mccann2$ei_measure[!is.na(filtered_mccann2$ei_measure)]
508 prop_table_f62 <- prop.table(table(ei_measure_data_f2)) * 100
509 prop_table_f62 <- round(prop_table_f62, 2)
510 prop_dff62 <- data.frame(Category = names(prop_table_f62), Percentage = prop_table_f62)
511 print(prop_dff62)

```

## Output 2:

```

> # Filter 2
> # Filter by test 1.1 and 3.1
>
> # Filter rows where ei_measure is either 1.1 or 3.1
> filtered_mccann2 <- filtered_mccann1[filtered_mccann1$ei_measure == 1.1 | filtered_mccann1$ei_measure == 3.1, ]
>
> # Get the number of rows and columns in the filtered data frame
> nrow(filtered_mccann2) # Number of rows
[1] 553
> ncol(filtered_mccann2) # Number of columns
[1] 20
> summary(filtered_mccann2) # Summary statistics
  study_id      sample_id      author      pub_year      yi      vi      pub_type      n      ed_level1
Min.   : 8.0   Min.   : 8.0   Length:553   Min.   :2000   Min.   :-0.58227   Min.   :0.00054   Min.   :0.0000   Min.   : 30.0   Min.   :2
1st Qu.:53.0   1st Qu.:55.0   Class :character   1st Qu.:2004   1st Qu.: 0.03025   1st Qu.:0.00269   1st Qu.:0.0000   1st Qu.: 85.0   1st Qu.:2
Median :98.0   Median :105.0   Mode  :character   Median :2007   Median : 0.13734   Median :0.00563   Median :0.0000   Median :150.0   Median :2
Mean   :107.6   Mean   :116.2   Mean   :2008   Mean   : 0.15342   Mean   :0.00750   Mean   :0.6259   Mean   :287.9   Mean   :2
3rd Qu.:163.0   3rd Qu.:177.0   3rd Qu.:2011   3rd Qu.: 0.27553   3rd Qu.:0.01010   3rd Qu.:1.0000   3rd Qu.:330.0   3rd Qu.:2
Max.   :246.0   Max.   :276.0   Max.   :2019   Max.   : 0.71828   Max.   :0.03421   Max.   :4.0000   Max.   :1771.0   Max.   :2
NA's   :136   NA's   :136   NA's   :136   NA's   :136   NA's   :136   NA's   :136   NA's   :136   NA's   :136
  ed_level2      country      perc_white      age      perc_female      ei_construct      ei_stream      ei_measure      subject
Min.   :1.000   Length:553   Min.   : 0.00   Min.   :17.74   Min.   : 0.00   Min.   :0.000   Min.   :1.000   Min.   :1.100   Min.   :0.0000
1st Qu.:2.000   Class :character   1st Qu.:52.00   1st Qu.:19.74   1st Qu.:62.02   1st Qu.:0.000   1st Qu.:1.000   1st Qu.:1.100   1st Qu.:0.0000
Median :2.000   Mode  :character   Median :67.80   Median :21.30   Median :68.70   Median :3.500   Median :3.000   Median :3.100   Median :0.0000
Mean   :1.803   Mean   :63.68   Mean   :21.88   Mean   :67.95   Mean   :3.702   Mean   :2.026   Mean   :2.126   Mean   :0.9248
3rd Qu.:2.000   3rd Qu.:89.25   3rd Qu.:23.00   3rd Qu.:73.62   3rd Qu.:6.000   3rd Qu.:3.000   3rd Qu.:3.100   3rd Qu.:3.100   3rd Qu.:0.0000
Max.   :2.000   Max.   :100.00   Max.   :40.00   Max.   :100.00   Max.   :9.000   Max.   :3.000   Max.   :3.100   Max.   :7.0000
NA's   :136   NA's   :333   NA's   :137   NA's   :174   NA's   :147   NA's   :136   NA's   :136   NA's   :141
  humanities      achievement_type
Min.   :0.0000   Min.   :0.0000
1st Qu.:0.0000   1st Qu.:0.0000
Median :0.0000   Median :0.0000
Mean   :0.1679   Mean   :0.1679
3rd Qu.:0.0000   3rd Qu.:0.0000
Max.   :1.0000   Max.   :1.0000
NA's   :136   NA's   :136
>
> # Quality Check: EI Measure
> length(filtered_mccann2$ei_measure)
[1] 553
> sum(is.na(filtered_mccann2$ei_measure))
[1] 136
> ei_measure_data_f2 <- filtered_mccann2$ei_measure[!is.na(filtered_mccann2$ei_measure)]
> prop_table_f62 <- prop.table(table(ei_measure_data_f2)) * 100
> prop_table_f62 <- round(prop_table_f62, 2)
> prop_dff62 <- data.frame(Category = names(prop_table_f62), Percentage = prop_table_f62)
> print(prop_dff62)
  Category Percentage.ei_measure_data_f2 Percentage.Freq
1      1.1             1.1             48.68
2      3.1             3.1             51.32

```

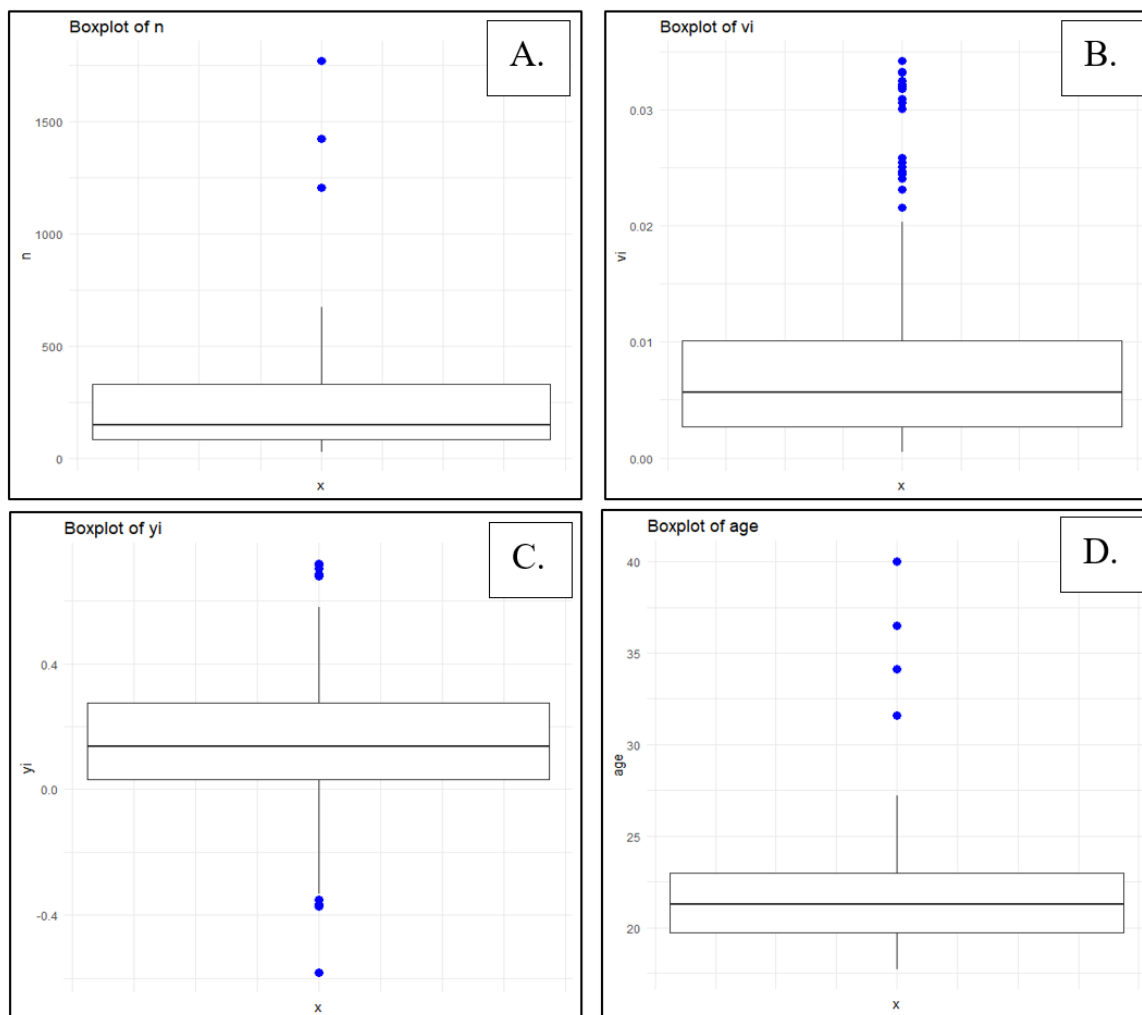
## Step 3: Descriptive statistics performed on the trimmed dataset to be used in the analysis.

**Method:** Descriptive statistics for the trimmed data were calculated using R software and represented in table format using Excel, where applicable, to provide a baseline for the data before it was trimmed. Descriptive statistics were calculated for variables explored in the original

data. The publication years range from 2000 to 2019, with a mean and median equal to 2007 and 2008 respectively. (Table 5; MacCann et.al., 2019). The effect size, represented by an  $r$

| Variable         | Min   | Max     | Median | Mean   | IQR     | Var       | SD     |
|------------------|-------|---------|--------|--------|---------|-----------|--------|
| pub_year (year)  | 2000  | 2019    | 2007   | 2008   | 7.00    | 16.87     | 4.11   |
| yi (r value)     | -0.58 | 0.72    | 0.14   | 0.15   | 0.25    | 0.04      | 0.20   |
| vi (SE^2)        | 0.00  | 0.03    | 0.01   | 0.01   | 0.01    | 0.00      | 0.01   |
| n (participants) | 30.00 | 1771.00 | 150.00 | 287.90 | 1741.00 | 112936.10 | 336.06 |
| perc_white (%)   | 0.00  | 100.00  | 67.80  | 63.68  | 100.00  | 668.84    | 25.86  |
| age (years)      | 17.74 | 40.00   | 21.30  | 21.88  | 22.26   | 10.58     | 3.25   |
| perc_female (%)  | 0.00  | 100.00  | 68.70  | 67.95  | 11.60   | 200.68    | 14.17  |

**Table 5:** Results of the descriptive statistics calculated for the Effect Size of the filtered MacCann 2019 data grouped by EI stream (1= ability EI and 3 = mixed EI). Calculations were performed in R. The two groups have similar mean  $r^2$  values of 0.15 (MacCann, 2019).



**Figure 2A-2D:** After the data trimming was completed, boxplots were created in R to evaluate the number of outliers in four quantitative variables:  $n$ ,  $vi$ ,  $yi$ , and  $age$ . The variables  $n$ ,  $vi$ , and  $age$  are right skewed pushing the median closer to the 3<sup>rd</sup> quartile. The variable  $yi$  is left skewed, pulling the median closer to the 1<sup>st</sup> quartile (MacCann, 2019).

correlation, ranged from approximately -0.58 to 0.72, with a median of 0.14 and a mean of 0.15 (Table 5; MacCann et.al., 2019). The variance and standard deviation for the effect size were 0.04 and 0.2, respectively (Table 5; MacCann et.al., 2019). This was unchanged from the original data. Boxplots were also produced to evaluate key variables for outliers (Figure 2A-2D).

**Conclusions:** The mean publication year was affected and dropped from 2010 to 2008 (MacCann et.al., 2019). The mean and median effect size was reduced to 0.14 and 0.15, respectively (MacCann et.al., 2019). Trimming the data only slightly affected the population statistics, reducing the median and mean to 150 and 288, respectively (MacCann et.al., 2019). The demographics also changed slightly. The mean percent proportion of the white population increased to 63.68% (MacCann et.al., 2019). The mean and median percent female population increased to 67.95% and 68.70%, respectively (MacCann et.al., 2019). The countries represented also decreased from 27 to 22 (MacCann et.al., 2019). The EI stream changed as follows: stream 1(48.68%), stream 2 (0%), and stream 3(51.32%) (MacCann et.al., 2019). The statistics for achievement type were not significantly changed. Data trimming minimally changed the descriptive statistics and demographics outlined for the original data. Data trimming by graduate education level resulted in an adequately sized working sample of  $n = 553$ . Descriptive statistics for the original data are viewable in Tables 4, as a comparison. The number of outliers was reduced by the second trimming step and the spread of the data is smaller.

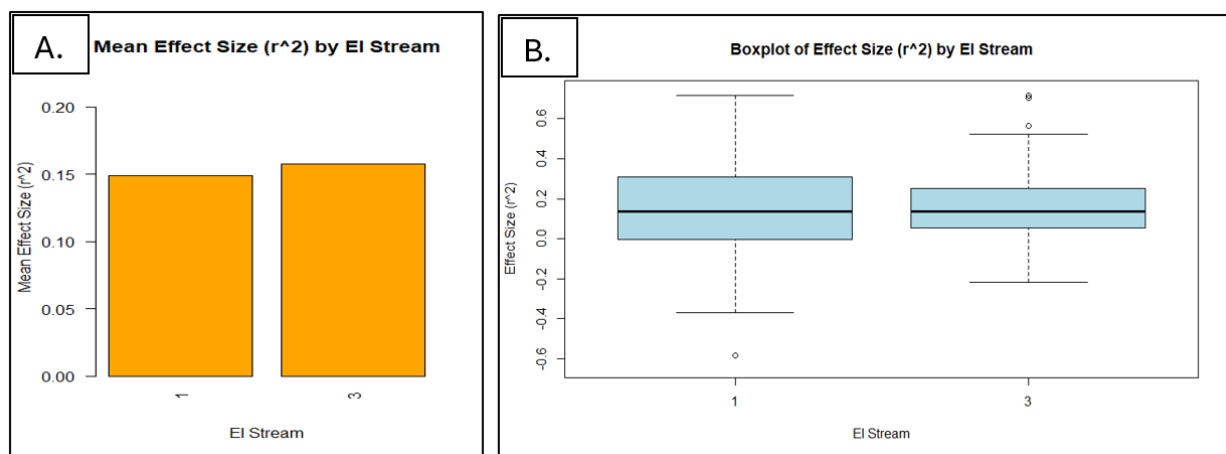
#### **Step 4: Exploratory data analysis to support the project proposal.**

After the second trimming procedure, the data was explored by grouping the data by EI stream (1= ability EI and 3 = mixed EI) then calculating descriptive statistics for the variable  $y_i$  (Effect Size =  $r^2$  value) for each of those groups (Table 6). The effect size represents the correlation coefficient, which measured the correlation between the results of the EI measures in each study and the corresponding academic achievement score in the form of a GPA or standardized test score. Based on the calculations performed in R, each group (1 and 3) has approximately the same mean  $r^2$  value of (0.15) (Table 6) (Figure 3A).  $R^2$  values fall between 0 and 1, with values closer to 1 being more significant. Since these mean  $r^2$  values are 0.15 and therefore closer to 0 there is either a very weak correlation between these variables or no relationship at all. One limitation of this dataset is the number of NaN values, which equaled 136 values within the trimmed data (Table 6). These values will be removed before running the

regression and branched decision tree model. The spread of the  $r^2$  values was then represented in a box plot, reflecting the minimum, maximum, mean and standard error (Figure 3B). The data seems to be relatively normally distributed. The  $r^2$  values for the ability EI measures are more spread out and have a larger standard error than the mixed EI values.

|   | ei_stream | mean_r2 | median_r2 | sd_r2 | min_r2 | max_r2 | count |
|---|-----------|---------|-----------|-------|--------|--------|-------|
|   | <dbl>     | <dbl>   | <dbl>     | <dbl> | <dbl>  | <dbl>  | <int> |
| 1 | 1         | 0.149   | 0.137     | 0.234 | -0.582 | 0.718  | 203   |
| 2 | 3         | 0.158   | 0.137     | 0.151 | -0.216 | 0.717  | 214   |
| 3 | NA NaN    | NA      | NA        | NA    | Inf    | -Inf   | 136   |

**Table 6:** Results of the descriptive statistics calculated for the Effect Size of the filtered MacCann 2019 data grouped by EI stream (1= ability EI and 3 = mixed EI). Calculations were performed in R. The two groups have similar mean  $r^2$  values of 0.15 (MacCann, 2019).



**Figure 3A-3B:** A. Visual representation of the mean effect size calculated in Table 4 for the filtered MacCann 2019 data grouped by EI stream (1= ability EI and 3 = mixed EI). Calculations were performed in R. The two groups have similar mean  $r^2$  values of 0.15. B. Boxplot of the  $r^2$  values within each group representing represented in a box plot, reflecting the minimum, maximum, mean and standard error. The mixed EI group has less spread than the ability EI group. Both groups seem relatively normally distributed (MacCann, 2019).

## B. Next Steps.

### Step 4: Statistical Analysis Methods.

The following analysis and modeling methods are presented, along with their relevance and usage, to explain whether emotional intelligence (EI) measures can serve as predictive markers for academic performance in graduate students,

## Modeling Methods

### A. Regression Analysis

Regression analysis will allow for a deeper examination of the relationship between EI and academic performance, controlling for variables such as demographic factors, previous academic performance and various others (Wooldridge, 2016). This method is critical for establishing a predictive model that can potentially identify how variations in Emotional Intelligence scores might forecast academic outcomes, providing valuable insights for graduate admissions, and highlights the unreliability of standardized test scores (Wooldridge, 2016).

### ***B. Branched Decision Tree***

For this study a branched decision tree will be utilized to address the hypothesis. Branched decision trees are machine learning models that use a structure like a decision tree (James et. al., 2013). The tree contains nodes which represent decision points (James et. al., 2013). From each node a branch extends, representing a possible outcome for that decision (James et. al., 2013). Since the correlation coefficients for this data have already been calculated in the data and converted into  $r^2$  values, called the effect size for each observation, the effect size will be used to calculate the decision tree. The two nodes to be evaluated will represent ability EI and mixed EI. All code used to implement this model was taken from the source outlined below.

Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. *An Introduction to Statistical Learning: With Applications in R*. New York: Springer, [2013] ©2013, 2013, <https://search.library.wisc.edu/catalog/9910207152902121>.

### ***C. Software.***

R for data analysis will be used for its powerful statistical capabilities, particularly for regression analysis, descriptive statistics, and use of a branched decision tree model. Excel will also be used for tables and calculations where appropriate.

### ***Step 5: Interpretation.***

Interpretation of the results from the above models will be used to evaluate the proposed research question. Further analysis will be performed if certain correlations stand out. Visualizations for the results of these models will be provided to further elucidate the interpretation of the outcomes of this study. The conclusions drawn from this data will provide

further evidence of the utility of EI measurements in predicting academic success and support further research into this burgeoning field.

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